

FITTRACK AI – PROJECT PHASE DOCUMENTATION

Phase 2.1: Problem - Solution Fit Template

Alignment Analysis Between User Pain Points, Manual Deficiencies, and Technical Backend Solutions

DOCUMENT REF	PHASE CATEGORY	TARGET SYSTEM	CORE PARADIGM
FIT-DOC-PH2.1-PSF	2. Project Design Phase	FitTrack AI RESTful Backend	MVC + Google Gemini AI

1. TARGET USER CONTEXT & PERSONA BACKGROUND

TARGET PROFILE: FITNESS ENTHUSIAST (CASE STUDY: RAHUL)

Background Context: Rahul is an active fitness enthusiast who exercises regularly across varied routines (Cardio, Running, Strength Training). He historically tracks exercise data manually using physical notebooks and standalone desktop spreadsheets. Despite strong dedication and consistency in workouts, manual logging methods hinder his long-term progress evaluation, routine organization, and objective guidance.

2. DOCUMENTED PROBLEM DEFINITION

The primary project documentation identifies 7 acute challenges inherent to unassisted, manual fitness tracking systems:

ID	DOCUMENTED PROBLEM	OPERATIONAL & BEHAVIORAL CONSEQUENCE
PR-01	Difficulty maintaining a centralized record	Physical logs and disparate spreadsheets become fragmented, misplaced, or difficult to reconcile across sessions.
PR-02	Inaccurate workout history from manual tracking	Human transcription mistakes, omission of session metrics, and inconsistent record formatting degrade historical integrity.
PR-03	No secure user authentication for records	Personal health, performance data, and exercise histories remain unprotected, unencrypted, and accessible to unauthorized parties.
PR-04	Unable to search previous workouts efficiently	Users cannot quickly query historical records by workout name, specific category, or specific calendar date.
PR-05	No personalized workout recommendations	Users lack professional guidance to dynamically tailor exercise regimens to their age, fitness goals, and experience levels.
PR-06	No insights regarding overall performance	Cumulative workout volume and caloric expenditure remain static figures without evaluative or motivational feedback.
PR-07	Time-consuming manual calculation	Manually calculating durations and caloric expenditure across routines wastes personal time and induces mathematical drift.

3. SCOPE GOVERNANCE & EXTENSION BOUNDARIES

Strict Scope Notice: In compliance with the primary project specification, the immediate solution addresses backend workout persistence, searching, JWT authentication, and Google Gemini AI insights. Features such as *Nutrition Tracking*, *BMI Calculation*, *Wearable Device Integration*, *Workout Reminders*, and *Fitness Analytics Dashboards* are documented as **Future Extensions** and are excluded from the current baseline fit.

4. DIRECT PROBLEM-TO-SOLUTION MAPPING MATRIX

The table below provides verifiable traceability demonstrating how each documented problem is directly resolved by specific architectural components, RESTful endpoints, and backend services implemented in FitTrack AI:

PROBLEM ID	ROOT PAIN POINT	FITTRACK AI TECHNICAL SOLUTION	SYSTEM COMPONENT / ENDPOINT
PR-01	Decentralized, scattered notebooks & files	Centralized NoSQL persistence in MongoDB workouts collection with schema integrity.	POST /api/workouts/ Mongoose Workout Model
PR-02	Inaccurate, non-standardized history	Strict Mongoose schema validation (required fields, enums, positive numbers, auto-timestamps).	models/Workout.js Centralized Error Handler
PR-03	Unsecured personal fitness logs	Encrypted password storage via bcrypt.js (salt rounds: 10) and stateless JWT verification middleware.	POST /api/auth/register POST /api/auth/login middleware/auth.js
PR-04	Inefficient manual record lookup	Optimized search API supporting query filtering by workout name, category enum, or workout date.	GET /search workoutController.js
PR-05	Lack of customized workout plans	Google Gemini AI service integration evaluating user age, fitness goals, and experience levels.	POST /api/ai/workout-recommendation geminiService.js
PR-06	Absence of performance evaluation	AI-driven fitness insights analyzing total workouts, average duration, and total calories burned.	POST /api/ai/fitness-insights geminiService.js
PR-07	Tedious manual metrics calculations	Automated aggregation and structured JSON responses for workout duration and caloric summaries.	workoutController.js JSON Response Payloads

5. VALUE PROPOSITION CANVAS SUMMARY

USER PAIN RELIEVERS (SYSTEM ALLEVIATIONS)	USER GAIN CREATORS (VALUE-ADD OUTCOMES)
• RELIEF Eliminates physical notebook entry and spreadsheet upkeep. • RELIEF Shields private data via JWT route middleware and bcrypt hashing. • RELIEF Prevents malformed inputs through automated schema constraints. • RELIEF Removes manual searching through rapid multi-parameter queries.	• GAIN Delivers intelligent AI-tailored weekly exercise routines. • GAIN Supplies motivational guidance, training tips, and safety advice. • GAIN Generates data-backed performance feedback on progressive overload. • GAIN Standardizes JSON responses across clients (Postman/Web/Mobile).